

MAE 184 Flight Simulation Techniques
Homework #4 Solution

Problem 4.1 (2 pts)

Write a function that will take four corner points of a 2D data table and estimate the value for any point that lies inside of the corner points. Demonstrate your function by estimating the value of thrust [N] for a point at Mach 0.44 and altitude 900 m.

Mach	Altitude = 0 m	Altitude = 2000 m
0.4	83487	68952
0.6	86328	71665

MATLAB:

```
function fxy = my_cfit(xg,yg,fg,x,y)
    X = [1 xg(1) yg(1) xg(1)*yg(1)
          1 xg(1) yg(2) xg(1)*yg(2)
          1 xg(2) yg(1) xg(2)*yg(1)
          1 xg(2) yg(2) xg(2)*yg(2)];
    b = [fg(1,1) fg(1,2) fg(2,1) fg(2,2)]';
    a = inv(X)*b; %coefficients
    fxy = a(1) + a(2)*x + a(3)*y + a(4)*x*y;
endfunction

function fxy = my_iterp(xg,yg,fg,x,y)
    fxy1 = ((xg(2)-x)*fg(1,1) + (x-xg(1))*fg(2,1))/(xg(2)-xg(1));
    fxy2 = ((xg(2)-x)*fg(1,2) + (x-xg(1))*fg(2,2))/(xg(2)-xg(1));
    fxy = ((yg(2)-y)*fxy1 + (y-yg(1))*fxy2)/(yg(2)-yg(1));
endfunction

% grid points
xp = [0.4 0.6]; % Mach
yp = [0.0 2000]; % altitude [m]
fp = [83487 68952; 86328 71665]; % thrust [N]

% test point
M = 0.44;
alt = 900;

ff = my_cfit(xp,yp,fp,M,alt) % 2d curve fit
ff2 = my_iterp(xp,yp,fp,M,alt) % 2d interpolation
ffm = interp2(xp,yp,fp',M,alt) % MATLAB function
```

ANSWER: Thrust = 77,500 N

Problem 4.2 (2 pts)

Create a time response plot to illustrate the dynamic effect of the T-38 pitch SAS modeled in JSBSim. The plot should show the pitch rate (q) response [y-axis, units: deg/sec] versus time [x-axis, units: sec]. The plot will show two responses, one with the pitch SAS enabled and a second with the pitch SAS disabled. Be sure to label each axis and use a legend to differentiate the two response curves. Submit your plot using an image format (BMP, PNG, JPG).

Aircraft: T38.xml (modified for MAE 184)

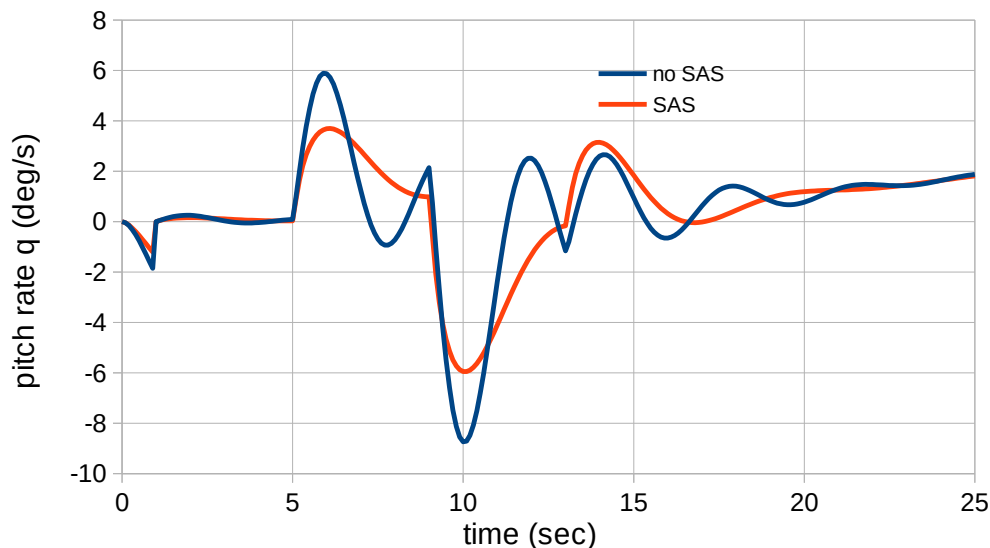
Flight Condition: Mach = 0.8, altitude = 40,000 ft

To disable the T-38 pitch SAS, change the `<gain>` value in the aircraft file "T38.xml".

```
<!-- set gain to 0.0 to disable pitch SAS -->  
<gain>1.0</gain>
```

Input: Normalized Elevator Command (fcs/elevator-cmd-norm) as generated by script "T38_elevator.xml" and shown below.

ANSWER:



Problem 4.3 (2 pts)

Estimate values for the two aerodynamic derivatives, C_{lp} and $C_{l\delta A}$, from the flight test data provided on Canvas. Both parameters should be adjusted to yield the best match to flight test data, but only C_{lp} needs to be reported in the answer.

Data File: T38_flight.csv

[time(sec) aileron(rad) roll_rate(rad/s) roll_angle(rad)]

Flight Condition:

altitude, $h = 40,000$ ft; Mach, $M = 0.86$

dynamic pressure, $q_{bar} = 200$ lbf/ft²

airspeed, $V = 830$ ft/s

Aircraft Geometry:

wingspan, $b = 25.5$ ft; wing area, $S = 170$ ft²

moment of inertia, $I_{xx} = 7850$ slug/ft³

Equations of Motion:

$$I_{xx} \frac{dp}{dt} = \bar{q} S b C_l \quad \frac{d\phi}{dt} = p$$

$$C_l = C_{lp} \left(\frac{pb}{2V} \right) + C_{l\delta A} \delta_A$$

$$p = \text{roll rate [rad/s]} \quad \phi = \text{roll attitude angle [rad]} \quad \delta_A = \text{aileron [rad]}$$

MATLAB

% Roll Dynamics

function xdot = EOM(x)

$I_{xx} = 7850.0$; % moment of inertia [slug/ft³]

$b = 25.5$; % wingspan [ft]

$S = 170.0$; % wing area [ft²]

$q_{bar} = 200$; % dyn press [lb/ft²]

$V = 830.0$; % airspeed [ft/s]

$C_{lp} = -0.3$; % roll damping

$C_{l\delta A} = 0.038$; % aileron effectiveness

$dA = x(1)$; % aileron [rad]

$p = x(2)$; % roll rate [rad/s]

$\phi = x(3)$; % roll attitude [rad]

$phat = 0.5 * p * b / V$; % normalized roll rate [rad]

$C_l = C_{lp} * phat + C_{l\delta A} * dA$; % roll moment coeff

$dp_dt = q_{bar} * S * b * C_l / I_{xx}$; % roll accel [rad/s²]

$dphi_dt = p$; % roll rate [rad/s]

```

        xdot = zeros(3,1);
        xdot(1) = 0.0; % aileron [sec]
        xdot(2) = dp_dt; % roll accel [rad/s2]
        xdot(3) = dphi_dt; % roll rate [rad/s]
    end

% RK4 Integration
function xnew = RK4(xold, h)
    r1 = EOM(xold);
    r2 = EOM(xold + 0.5*h*r1);
    r3 = EOM(xold + 0.5*h*r2);
    r4 = EOM(xold + h*r3);
    xnew = xold + (h/6.0)*(r1 + 2.0*r2 + 2.0*r3 + r4);
end

% read data file
m = csvread("T38_aileron.csv");
tt = m(:,1); % time [s]
dA = m(:,2); % aileron [rad]
p = m(:,3); % roll rate [rad/s]
phi = m(:,4); % roll angle [rad]

% simulation setup
dt = tt(2) - tt(1); % integration time step [s]

% set initial conditions
x = [dA(1); % start time [s]
p(1); % roll rate [rad/s]
phi(1)]; % roll attitude [rad]

% run simulation
for i = 1:length(tt)
    x(1) = dA(i); % aileron data as input
    xs(:,i) = x; % store state vector in columns
    x = RK4(x, dt); % RK4 step
end

% extract variables for plotting
ps = (180.0/pi)*xs(2, :); % roll rate [deg/s]
phis = (180/pi)*xs(3,:); % roll attitude [deg]

% compute match
p_err = p - xs(2,:);
phi_err = phi - xs(3,:);
JJ = 0.5*(p_err'*p_err + phi_err'*phi_err)

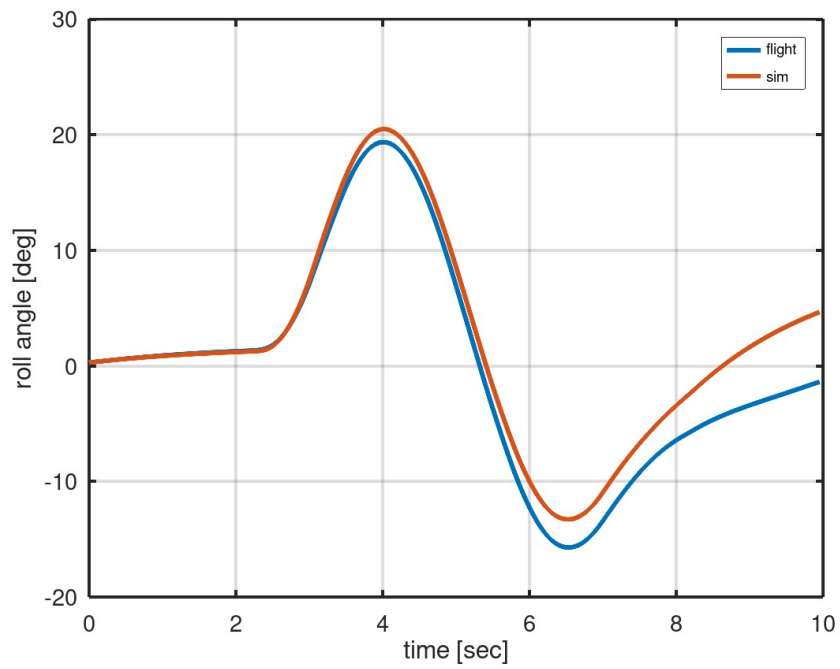
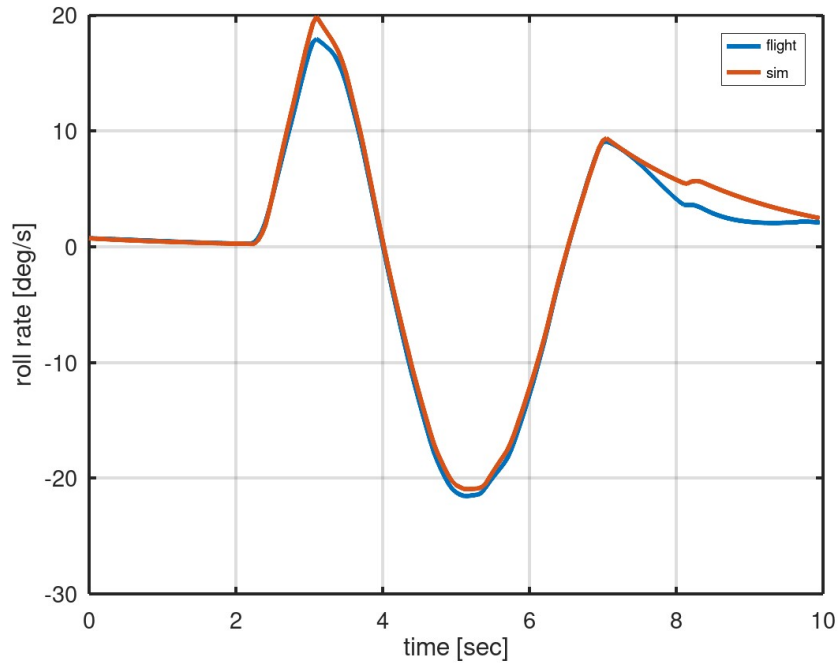
figure(1)
plot(tt,p*180/pi,"linewidth",3.0,tt,ps,"linewidth",3.0)
xlabel("time [sec]")
ylabel("roll rate [deg/s]")
legend("flight","sim")
set(gca,"linewidth",2.0,"fontsize",14)
grid on

figure(2)
plot(tt,phi*180/pi,"linewidth",3.0,tt,phis,"linewidth",3.0)
xlabel("time [sec]")
ylabel("roll angle [deg]")

```

```
legend("flight","sim")
set(gca,"linewidth",2.0,"fontsize",14)
grid on
```

ANSWER: $C_{lp} = -0.3$, $C_{l\dot{\alpha}} = 0.038$ ($C_{l\dot{\alpha}}$ not graded)



Problem 4.4 (2 pts)

Find the two possible body positions, defined by symmetric positive and negative values of theta (tilt angle of leg L3), that result when solving the 'forward problem' for the 2D Planar Motion System described in Lecture 22 (Motion Systems).

$$p1 = p2 = p3 = \sqrt{5}$$

$$L1 = 2, L2 = L3 = \sqrt{2}$$

$$\gamma = \pi/2, x1 = 4, x2 = 0, y2 = 4$$

MATLAB

```
function dd = planar(theta)
    gam = pi/2.0;
    p1 = sqrt(5.0);
    p2 = p1;
    p3 = p1;
    L1 = 2.0;
    L2 = sqrt(2.0);
    L3 = L2;
    x1 = 4.0;
    x2 = 0.0;
    y2 = 4.0;
    A2 = L3*cos(theta) - x1;
    B2 = L3*sin(theta);
    A3 = L2*cos(theta + gam) - x2;
    B3 = L2*sin(theta + gam) - y2;
    N1 = B3*(p2*p2-p1*p1-A2*A2-B2*B2) - B2*(p3*p3-p1*p1-A3*A3-B3*B3);
    N2 = A2*(p3*p3-p1*p1-A3*A3-B3*B3) - A3*(p2*p2-p1*p1-A2*A2-B2*B2);
    D = 2.0*(A2*B3 - B2*A3);
    dd = N1*N1 + N2*N2 - p1*p1*D*D;
endfunction

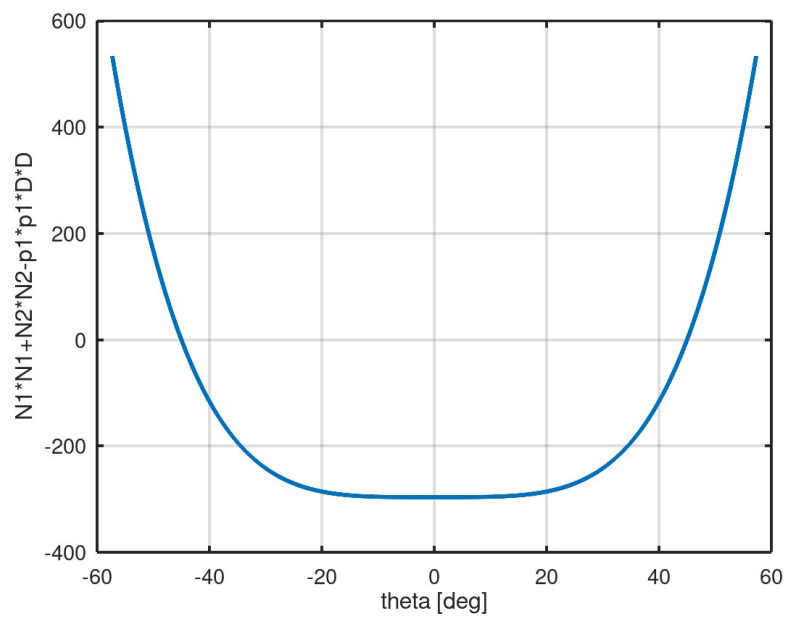
% find solutions in plot
thetar = [-1.0:.01:1.0];
for i =1:length(thetar)
    ddr(i) = planar(thetar(i));
end

% check solusions
theta1 = -45.0
dd1 = planar(theta1*pi/180)
theta2 = 45.0
dd2 = planar(theta2*pi/180)

figure(1)
plot(thetar*180/pi,ddr,"linewidth",3.0)
xlabel("theta [deg]")
ylabel("N1*N1+N2*N2-p1*p1*D*D")
set(gca,"linewidth",2.0,"fontsize",14)
grid on
```

ANSWER:

```
theta1 = -45  
dd1 = -2.2737e-13  
theta2 = 45  
dd2 = -2.2737e-13
```



Problem 4.5 (2 pts)

Estimate the elevator hinge moment HM for a Cessna 182 flying at a level-flight trim condition where the dynamic pressure is 30 lb/ft^2 , angle-of-attack is $+1.8 \text{ deg}$, and the elevator is deflected $+2.0 \text{ deg}$.

$Ch_0 = 0$, $Ch_{\alpha} = -0.06 \text{ (1/rad)}$, $Ch_{\delta E} = -0.6 \text{ (1/rad)}$

horiz. tail area = 38.5 ft^2
horiz. tail chord = 3.4 ft

- | | |
|------------------------|--------------------------|
| a. -3 ft.lb | c. -170 ft.lb |
| b. -90 ft.lb | d. -5140 ft.lb |

MATLAB:

```
% C182 geometry and hinge moment coefficients
Sh = 38.5; # horiz tail area [ft2]
cbarh = 3.4; # horiz tail chord [ft]
Ch0 = 0.0;
Ch_alpha = -0.06; # [1/rad]
Ch_deltaE = -0.6; # [1/rad]

% flight condition
qbar = 30.0; # dynamic pressure [lb/ft2]
alpha = 1.8*pi/180.0; # angle of attack [rad]
deltaE = 2.0*pi/180.0; # elevator deflection [rad]

Ch = Ch0 + Ch_alpha*alpha + Ch_deltaE*deltaE;

HM = qbar*Sh*cbarh*Ch
```

ANSWER: $HM = -89.6 \text{ ft.lb}$